Bayesian Discrete Choice Models

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Discrete Choice Analysis

- Discrete choice analysis
  - is a technique that determines which attributes of a product are most important to customers
  - is based on the assumption that consumers make purchase decisions by considering multiple attributes of a product simultaneously
  - is sometimes referred to as “choice-based conjoint analysis”
  - has become the most widely used conjoint technique in the world.

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Discrete Choice Analysis

- In marketing research, discrete choice analysis is used to
  - configure a new product or service
  - re-configure an existing product or service
  - better understand what drives customer preferences in the marketplace
  - set prices for maximum profitability.

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Discrete Choice Analysis

- subjects are presented with sets of hypothetical products and are asked to choose the most preferred hypothetical product from each set
- the aim of the analysis is to discover which attribute levels affect their choices
- the subjects should have an interest in, use, or purchase the types of products in the discrete choice study.

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Choice and Survival Models – Classical View

- In survival models fit in PROC PHREG, the data consist of time until an event occurs or time until censored (subject did not experience the event).
- In discrete choice models fit in PROC PHREG, the first choice (the event) occurs at time 1. All the alternatives that were not chosen are censored at time 2.
- The survival and discrete choice models are the same.

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BCHOICE Procedure

- PROC MCMC is the general-purpose procedure where BCHOICE is specifically designed for discrete choice models
- Syntax and diagnostics are similar between the two
  - PREDDIST
  - RANDOM
- BCHOICE has a CLASS statement

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Simple Logit Model Example

- 10 subjects presented with 8 different chocolate candies choosing one

- Dark Chocolate vs Milk Chocolate
- Soft Center vs Chewy Center
- Nuts vs No Nuts

- Multinomial logit model

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Logit Model Example with Random Effects

- Market Demand for kitchen trash cans
- 104 people enrolled assigned with 10 choice tasks to each
- Participants stated their preference between two types of trash cans

- 4 attributes each with two levels
  - Touchless opening
  - Material (steel/plastic)
  - Automatic bag replacement
  - Price ($40/$80)

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